

Evaluating Computational offloading with Edge Computing for IoT devices

Following are the evaluation metrics we are going to use for evaluating our research and development project.

1. Impact of the arrival time of offloaded tasks.

This should be evaluated as the arrival rate of tasks to the MEC server affects the offloading cost and the length of the queue of tasks at MEC. So we can take different arrival rates and check how that affects the offloading cost.

2. Evaluating our algorithm to offload with existing algorithms.

We can evaluate our algorithm with another two types of algorithms. One is a randomized algorithm where, offloads the tasks requests to the MEC or back end cloud randomly. Other type is a MEC only algorithm which offloads all the task requests to the MEC for processing. Overall offloading cost of different algorithms and Queue length of different algorithms can be then compared, to measure the performance of our proposed algorithm.

3. Energy Consumption of the used method

Real time IoT devices must consider their energy consumption as it affects their performance. That could be measured by the time taken for the battery to lose its charge to a certain level. This can be measured with different type of algorithms.

4. Comparison time taken between with and without offloading feature

We should evaluate the given worst case process and best case process in the IoT device with and without the offloading method. Simply what it means is we should know that the process really need the resource rich device to compute something. If the worst case process can be done with neglectable time delay without the offloading there is no purpose in doing offloading for the process.

$\text{Time taken}(X) = \text{offloading time} + \text{process time} + \text{result sending/receiving}$

Time "X" for the worst case < Time taken for the worst case of isolated IoT device. --- **B**

Time "X" for the best case < Time taken for the best case of isolated IoT device ----- **A**

By considering the "A" and "B" gives true or false we can have an idea of the importance of this research. So we can evaluate our research need for the given IoT device in this way.

5. Testing the developed platform with the different IoT device base technologies like Computer vision, Augmented reality etc.

We are going to evaluate our offloading platform with different kind of technologies because sometimes developed algorithm and platform may only support limit technologies.

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